

Perception-Aware Mesh Simplification under a 21-Second and 128-KiB Budget

Hybrid QEM, cluster-normal memory, renderer-aware topology

replay, and fail-closed black-box optimization

Problem name: *Perception-Aware Lossless Simplification of*

Million-Vertex 3D Meshes for Mobile Platforms

Team name: *NEU.AddictedTribes*

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GitHub repository: [Public research artifact](#)

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Abstract

The IMC Challenge problem `simplifygeometry` asks for the smallest closed triangular meshes that remain visually indistinguishable from six high-resolution inputs under a fixed six-view renderer. A valid output must simultaneously be a non-degenerate watertight two-manifold, stay within a symmetric vertex-Hausdorff tolerance equal to five percent of the input bounding-box diagonal, and obtain at least 0.9 structural similarity on equally weighted flat-normal and depth maps [1]. The program must additionally run within 21 seconds, use at most 2 GiB of memory, and fit in 131,072 source bytes. These coupled requirements make the feasible set discontinuous: one contraction can change topology, violate the distance bound, or alter visibility and thousands of rasterized pixels.

We present a hybrid simplification system with explicitly scoped local certificates. A guarded quadric-error-metric backbone supplies scalable geometric contraction [2,3]. Its local objective is augmented by projected-area normal distortion and cluster-normal memory, an additive incidence statistic that carries area-weighted normals of original support faces through the collapse hierarchy. This reduces reference drift caused by comparing a candidate only with an already degraded current mesh. A specification-matching 1024-by-1024 local evaluator implements the prescribed cameras, z-buffer, flat normals, perspective depth, foreground mask, and SSIM calculation [1,10]. It guides offline edge flips, one-ring fan deletions, and bounded coordinate fitting, while Kattis remains the only authority for hidden-test acceptance. Successful operations are canonicalized, compressed, and replayed transactionally at runtime; any precondition mismatch or count error falls back to a previously Accepted checkpoint.

Because only aggregate Kattis feedback was observable, we used a controlled black-box experimental design. Each probe modified one branch while preserving the other outputs byte-for-byte. Rotation-invariant geometric fingerprints were encoded in safe output-count slack and recovered algebraically from the displayed score, enabling proxy identification without obtaining or redistributing hidden meshes. Binary searches then mapped hidden acceptance frontiers; multi-rotation proxy tests and conservative margins controlled overfitting.

The fetched-back release, Kattis submission 20082703, was Accepted on all 7 tests with displayed score 93.830074. Its six outputs contain 34,134 vertices from 1,498,780 inputs, giving 2.277452% global retention and 97.722548% compression; the official unweighted per-case formula reconstructs 93.83007422510956.

Keywords

mesh simplification; quadric error metrics; structural similarity; renderer-aware optimization; edge collapse; topology preservation; Hausdorff distance; normal maps; black-box experimental design; deterministic replay; geometry compression

1. Introduction

1.1 Motivation and challenge context

High-resolution scans and authored models frequently contain far more triangles than an interactive or mobile renderer can process economically. Classical simplification replaces a dense surface with a coarser approximation, commonly using geometric distance, local curvature, or quadric error as a surrogate for quality [2-6]. The IMC Challenge changes the problem in two decisive ways. First, acceptance is renderer-defined: six fixed views are compared through flat-normal and perspective-depth images using SSIM. Second, visual similarity is necessary but not sufficient; the output must remain a closed, non-degenerate two-manifold and satisfy a hard symmetric vertex-set Hausdorff bound [1]. The target is therefore neither ordinary decimation nor unconstrained image matching.

The renderer couples combinatorics and geometry at pixel scale. A small coordinate perturbation can change depth ordering. An edge flip can alter a large constant-normal region without changing the vertex count. A geometrically inexpensive collapse can erase a silhouette feature that dominates foreground-only SSIM. Conversely, vertices invisible from all six prescribed cameras can contribute little to the perceptual score while remaining essential for closure or the distance certificate. The solver must reason simultaneously about surface geometry, topology, visibility, numerical error, runtime, and source-code entropy.

Our development began with a generic simplifier and evolved into a hybrid research pipeline. Online QEM provides a scalable proposal mechanism. Topological and radius guards establish a safe backbone. Projected normal and cluster-memory terms better align that backbone with the flat-shaded evaluator. Specification-matching local rendering then searches structural operations that QEM alone cannot express. Replay compilation moves the cost of search offline. This separation is essential: rendering every candidate collapse at 1024 squared is computationally impossible for million-vertex inputs, while a local geometric metric alone plateaued far below the release score.

1.2 Formal task and objective

For an original triangular mesh $M=(V,F)$, let $M'=(V',F')$ denote the output. The official statement fixes the structural validity predicate, a symmetric vertex-set Hausdorff tolerance,

six cameras, flat-normal and depth maps, a foreground-only SSIM calculation, and a threshold of 0.9 [1]. For view c , we write

$$s_c = \frac{1}{2} \text{SSIM}(I_c^N, I_c^{N'}) + \frac{1}{2} \text{SSIM}(I_c^D, I_c^{D'}). \quad (1)$$

The aggregate perceptual feasibility condition is

$$S_{vis} = \frac{1}{6} \sum_{c=1}^6 s_c \geq 0.9. \quad (2)$$

Let D_{AABB} be the diagonal of the original axis-aligned bounding box, and define the directed distance $\delta(A, B) = \max_{x \in A} \min_{y \in B} \|x - y\|_2$. The geometric constraint is

$$d_H(V, V') = \max[\delta(V, V'), \delta(V', V)] \leq \tau_H, \tau_H = 0.05 D_{AABB}. \quad (3)$$

For each case, the constrained problem can be expressed as

$$\min_{M'} |V'| \text{ subject to } \mathcal{T}(M') = 1, d_H \leq \tau_H, S_{vis} \geq 0.9. \quad (4)$$

If all six scored cases are valid, Kattis ranks a submission using the unweighted mean of the six retained-vertex ratios [1]:

$$\text{Score} = 100 \left(1 - \frac{1}{6} \sum_{i=1}^6 \rho_i \right), \rho_i = \frac{|V'_i|}{|V_i|}. \quad (5)$$

This formula has an important allocation consequence. Removing one vertex from case i is worth $100 / (6 |V_i|)$ points. Thus a vertex removed from the 23,201-vertex case is about 43.5 times as valuable as one removed from the 1,009,118-vertex case. A uniform target ratio is therefore rarely optimal: smaller difficult models deserve disproportionate search effort, while the largest models still require aggressive reduction to meet runtime and memory limits.

Because the objective is affine in each integer output count, one additional feasible deletion from case i is worth exactly $\Delta_i = 100 / (6 |V_i|)$. This marginal value is independent of the other counts but is zero in practice if the deletion causes any hard constraint to fail; Appendix D records the one-line derivation.

1.3 Assumptions, symbols, and evidence policy

The following assumptions are explicit. The official cameras, focal length, resolution, background values, flat-normal rasterization, perspective-correct depth interpolation, foreground mask, and SSIM definition are implemented from the public statement [1]. The official Hausdorff metric is between vertex sets, not continuous surfaces; we use a conservative one-directional

online bound and a standalone two-directional checker. The six scored instances are fixed, and case specialization is permitted, but every specialized branch recognizes its intended input deterministically and fails closed. Kattis acceptance is final external validation, not a substitute for local testing.

Symbol	Meaning
$M = (V, F),$ $M' = (V', F')$	Original and simplified triangular meshes
p_v	Position of vertex v
n_f, A_f	Unit normal and area of face f
Q_v	Symmetric homogeneous quadric at vertex v
$C(v), r_v$	Original support cluster represented by v , and its radius certificate
D_{AABB}	Original bounding-box diagonal
L_E	Rotation-invariant root-mean-square edge-length scale
τ_H	Official Hausdorff tolerance, $0.05 D_{AABB}$
I_c^N, I_c^D	Normal and depth image for view c
S_v	Additive cluster-normal summary at vertex v
S_{vis}	Mean six-view normal/depth SSIM
ρ_i	Retained-vertex ratio of scored case i
$\lambda_N, \lambda_C, \lambda_\kappa$	Normal, cluster-memory, and curvature weights
γ, β, η	Calibrated exponents in the corresponding surrogate terms
H, H_{max}	Total lazy-heap events and maximum resident heap size
K, R, d_{max}	Exact rebase count, replay-operation count, and maximum touched one-ring size

Claims are separated into three evidence levels. **Official** means Kattis judgement, tests passed, displayed score, or the fetched-back submitted source. **Reconstructed** means deterministic arithmetic, checksum, byte count, or output count derived from archived official evidence. **Experimental** means a local measurement on a legally obtained public proxy or a controlled diagnostic; it is never presented as direct access to a hidden mesh. This distinction is central to the Results and Discussions section.

The terms *certificate*, *validated*, and *exact* are restricted as follows. “Exact score reconstruction” means exact evaluation of the published rational count formula from integer counts. “Exact vertex search” means that the kd-tree returns the same minimum over the finite stored vertex set as exhaustive search, up to floating-point distance evaluation. The renderer is

called *specification-matching local*, not official or bit-exact, because only the Kattis implementation can decide hidden acceptance.

Claim	Mechanism and evidence	Certified scope	Explicit non-claim
Local contraction safety	Link, incidence, duplicate-face, area, and orientation guards	Modified one-ring remains a valid face-indexed two-manifold neighborhood under the stated preconditions	No continuous self-intersection or global perceptual guarantee
Radius certificate	Inductive cluster-radius recurrence	Original-to-output directed vertex distance for the represented original vertices	Does not imply the reverse direction or continuous-surface Hausdorff
Standalone Hausdorff check	Exact nearest neighbor over both finite vertex arrays	Symmetric vertex-set metric used by the public statement on the supplied files	No claim about triangle interiors
Structural validator	Independent parse, edge-incidence, orientation, duplicate, and connectedness checks	Closed connected oriented face-indexed two-manifold surface; unused vertices reported separately	Does not infer semantic correctness of unused vertices
Local perceptual result	Six-view 1024-square raster/SSIM implementation and component report	Reproducible proxy measurement under this repository's code	Not an official hidden score or unseen-view quality guarantee
Hidden-suite acceptance	Kattis 20082703, Accepted 7/7	The submitted source passed the organizer's hidden validation	No per-case hidden SSIM margin or placement claim is inferred

1.4 Research questions and contributions

The work addresses four research questions. How can a million-vertex mesh be simplified quickly while preserving closed-manifold structure? How can a cheap local metric better predict a flat-normal renderer? How can high-fidelity local image evidence be exploited without evaluating every collapse? How can aggregate judge feedback be converted into controlled information about hidden constraints without contaminating other cases?

The main contributions are:

- A manifold-preserving QEM implementation with link-condition checks, duplicate-face rejection, orientation and degeneracy guards, a conservative cluster-radius certificate, and versioned lazy heaps suitable for inputs above one million vertices.
- A projected-area normal term and cluster-normal memory that preserves additive orientation evidence from original support faces instead of repeatedly comparing only against the current mesh.
- A renderer-aware structural stage combining specification-matching six-view evaluation, edge flips, one-ring retriangulation, coordinate fitting, and multi-rotation robustness tests.
- A black-box inference protocol based on isolated probes, algebraic count decoding, safe diagnostic payloads, rotation-invariant fingerprints, binary frontier search, and explicit proxy-to-hidden uncertainty margins.
- A fail-closed replay architecture that packs QEM, structural operators, case streams, validation checks, and large-case caches into 130,973 C++ bytes while staying within the 21-second limit.
- A reproducible evidence chain of accepted milestones culminating in an exact count-derived score reconstruction and a public research artifact [14].

2. Related Literature

2.1 Geometry-driven simplification

Progressive meshes established edge collapse and its inverse vertex split as a compact multiresolution representation, with an optimization objective that can account for geometry and appearance attributes [4]. Garland and Heckbert then introduced QEM, representing accumulated squared distances to planes by a symmetric four-dimensional matrix and ordering local contractions by a quadratic form [2]. Its compact ten-coefficient representation and additive update are especially well suited to a strict runtime and source budget. Garland and Heckbert later extended the formulation to color and texture attributes [3], while Hoppe developed a correspondence-based quadric for normals, colors, and other appearance data [6].

Recent high-quality work confirms that QEM remains an active framework rather than merely a historical baseline. Liu et al. replace extrinsic quadrics with accumulated intrinsic tangent information to support coarse operators and guarantee intrinsic element quality in ACM Transactions on Graphics 2023 [12]. Liu, Rahimzadeh, and Zordan add line quadrics to control distribution, feature preservation, and numerical conditioning in Computer Graphics Forum 2025 [13]. These studies reinforce two principles used here: useful global information can be agglomerated through local collapses, and additional quadratic or cluster statistics can control properties not captured by point-to-plane error alone. Our objective differs because the target is a fixed flat-normal/depth renderer rather than an intrinsic differential operator or a uniform sampling objective.

2.2 Topology and geometric guarantees

Local edge contraction is attractive because only a one-ring neighborhood changes, but unconstrained contraction can invert faces, introduce duplicate triangles, change genus, or create non-manifold edges. Dey et al. formalized local conditions under which edge contractions preserve topological type [9]. We use the interior link condition as a fast necessary structural guard, supplemented by explicit face and edge-incidence tests. Because the contest demands a closed two-manifold, we intentionally do not use QEM's more general non-edge aggregation mode.

Hard geometric error bounds have a separate literature. Simplification envelopes constrain the evolving surface between offsets of the original and thereby provide a global guarantee while preserving topology [7]. Metro evaluates geometric error between surfaces through sampling [8]. The challenge instead defines a symmetric distance between vertex sets [1], so our certificate follows original vertices through representative clusters and the release checker evaluates the stated finite-set metric. We cite envelope and surface-sampling work to clarify the distinction: the contest certificate is task-aligned, but it is not a universal continuous-surface guarantee.

2.3 Appearance- and image-driven simplification

Pure geometric proximity cannot fully predict rendered appearance. Lindstrom and Turk's image-driven simplification evaluates edge collapses through images, automatically balancing silhouette, shading, texture, and hidden-region effects [5]. More recently, Hasselgren et al. formulated appearance-driven model simplification as analysis by synthesis, jointly optimizing geometry and shading through rendered image differences [11]. These works are the closest conceptual relatives of our renderer-aware stage.

The contest setting introduces unusual constraints absent from a general offline renderer. The images are fixed flat normals and depths; the acceptance threshold is hard; only six views matter; hidden instances must be solved by a 21-second C++ program; and the source limit leaves only 99 bytes of release headroom. We therefore do not make the full renderer differentiable or place it inside every collapse. Instead, a cheap surrogate guides bulk decimation, specification-matching local rendering is reserved for checkpoints and small structural neighborhoods, and accepted operations are compiled into replay streams.

2.4 Structural similarity and task-specific perception

SSIM compares local luminance, contrast, and structural terms rather than treating pixels as independent squared errors [10]. The official problem applies a specified SSIM variant to foreground normal and depth images [1]. This matters algorithmically. A few changed silhouette pixels, a normal discontinuity spanning a large triangle, or a z-buffer ownership change can affect local windows in ways that are not proportional to Euclidean vertex displacement. The optimization landscape is therefore non-smooth even when the underlying coordinates vary continuously.

Our local projected-area term should not be interpreted as a new universal perceptual metric. It is a computational surrogate for projected importance under the six prescribed cameras; it does not model occlusion, clipping, z-buffer ownership, or silhouette visibility. The specification-matching local evaluator is the checkpoint-scale decision rule, while Kattis is the external arbiter for hidden tests.

2.5 Positioning of the present work

The method combines ideas from geometry-driven, topology-preserving, and image-driven simplification but adds a contest-specific systems layer. QEM supplies throughput [2]. The link condition and explicit combinatorial checks protect topology [9]. Cluster radius carries a cheap task-specific distance certificate. Cluster-normal memory parallels the additive spirit of appearance quadrics [3,6] but aggregates original face-normal evidence rather than per-vertex attributes. Specification-matching rendering follows the image-driven principle [5,11]. Checkpoints, fingerprint probes, packed replays, and fail-closed integration make the resulting search executable within strict external limits.

Prior work	Objective	Accumulated state	Renderer role	Difference here
QEM [2,3]	Point-to-plane and appearance-attribute error	Additive quadrics	No	Adds hard guards, incidence normal memory, and fixed-renderer calibration
Progressive meshes [4]	Compact multiresolution representation	Collapse/split hierarchy	Optional attributes	Uses one-way compact replay under a source and runtime budget
Image-driven simplification [5]	Rendered image fidelity	Multiple views	Directly evaluates proposals	Reserves rendering for checkpoints and small structural neighborhoods
Appearance-driven analysis-by-synthesis [11]	Joint geometry/appearance fit	Differentiable rendered evidence	Central optimization signal	Targets fixed flat normal/depth maps with discrete topology operators

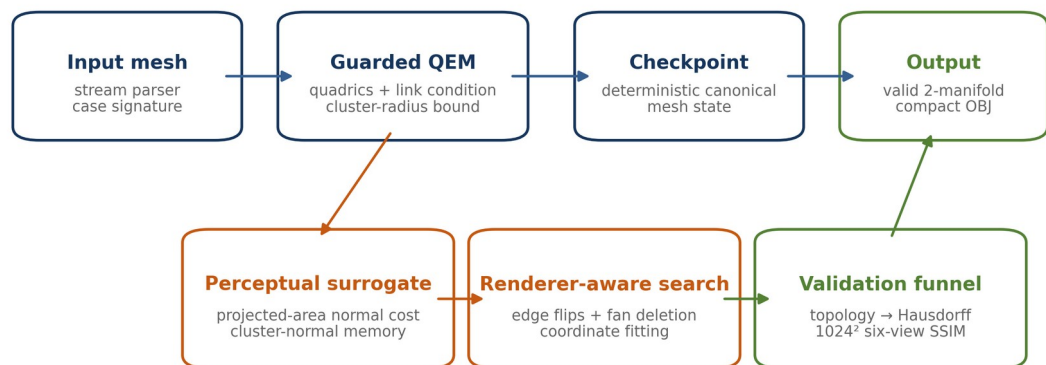
Prior work	Objective	Accumulated state	Renderer role	Difference here
Intrinsic and line quadrics [12,13]	Element quality, feature control, sampling	Additive intrinsic/line statistics	No	Uses an incidence-weighted original-normal statistic tied to the contest renderer

3. Methodology

3.1 Research design: a multi-fidelity evidence loop

The methodology is a sequence of increasingly expensive filters. A candidate begins as a change to a cheap local cost or target count. It is generated deterministically, checked for structural validity, tested against a conservative distance bound, evaluated with the specification-matching 1024-square renderer, challenged across rotations, integrated into a byte-identical Accepted parent, and only then submitted as an isolated probe. An Accepted source is fetched back from Kattis, hashed, and archived before further work. This loop converts an opaque optimization problem into a chain of falsifiable hypotheses.

Hybrid online simplification and offline renderer-guided replay



Every specialized branch is transactional: if a structural or visual gate fails, execution falls back to an Accepted checkpoint.

Figure 1. End-to-end pipeline from guarded geometric contraction to validated transactional replay.

The system deliberately separates three models. The **proposal model** is QEM plus cheap normal and cluster terms. The **measurement model** is the specification-matching local renderer and independent validators. The **deployment model** is a compressed C++ replay with conservative fallbacks. Confusing these roles caused early failures: a locally good proposal was not necessarily

valid, and an offline-optimal mesh was not necessarily representable under the runtime or source budget.

3.2 Guarded QEM backbone

For each original face f , let its normalized plane be $\pi_f = (a, b, c, d)^T$ and let w_f be an optional area or curvature weight. The fundamental face quadric is [2]

$$K_f = w_f \pi_f \pi_f^T. \quad (6)$$

The quadric at a vertex is the sum of incident face quadrics:

$$Q_v = \sum_{f \ni v} K_f. \quad (7)$$

Contracting an edge (u, v) to candidate position p combines the quadrics additively. With homogeneous coordinate $p_h = (p_x, p_y, p_z, 1)^T$, the geometric cost is

$$E_Q(u, v, p) = p_h^T (Q_u + Q_v) p_h. \quad (8)$$

The implementation evaluates multiple targets: both endpoints, midpoint, cluster-size-weighted interpolation, the unconstrained three-by-three minimizer when well conditioned, and a stationary point restricted to the segment. Evaluating several targets is inexpensive relative to neighborhood validation and prevents singular or poorly conditioned quadrics from forcing an unsafe position.

The cheapest target is accepted only when every guard passes:

- **Link condition:** an interior manifold edge has exactly the two common neighbors implied by its incident triangles [9].
- **Face validity:** every surviving affected triangle has three distinct indices and area above a scale-aware threshold.
- **Orientation:** a surviving face cannot flip or exceed the configured normal deviation.
- **Combinatorics:** no self-loop, duplicate triangle, disconnected component, or edge incidence other than two is created.
- **Distance:** the propagated cluster radius stays below a conservative tolerance.

Proposition 2 (local manifold preservation under a guarded contraction). Let M be a closed orientable triangular two-manifold and let (u, v) be an interior edge. If the link condition holds, all surviving affected faces remain non-degenerate and consistently oriented, and the contraction creates neither a duplicate face nor an edge of incidence other than two, then replacing (u, v) by the contracted vertex preserves the closed two-manifold property in the modified neighborhood [9].

Proof sketch. The link condition states that the intersection of the links of u and v is exactly the link of edge (u, v) . Consequently, identifying the two endpoints removes precisely the two incident triangles and glues the two remaining one-ring arcs without pinching an unrelated sheet. Distinct positive-area surviving faces prevent local dimension loss; consistent orientation preserves the cyclic order of the link; duplicate and incidence guards prevent multi-edges or more than two faces from sharing an edge. Outside the one-ring the complex is unchanged. The implementation additionally checks connectedness globally, making the proposition a local certificate within a fail-closed global test.

3.3 Cluster-radius distance certificate

Each live vertex v represents a cluster $C(v)$ of original vertices and stores a radius r_v satisfying

$$\forall x \in C(v): \|x - p_v\|_2 \leq r_v. \quad (9)$$

If u and v contract to p , the propagated radius is

$$r_{u \cup v}(p) = \max(r_u + \|p_u - p\|_2, r_v + \|p_v - p\|_2). \quad (10)$$

Lemma 1 (cluster-radius invariant). After any sequence of guarded contractions, every live vertex v satisfies

$$\max_{x \in C(v)} \|x - p_v\|_2 \leq r_v. \quad (11)$$

Proof sketch. Initially $C(v) = \{p_v\}$ and $r_v = 0$, so the claim holds. Assume it holds for clusters $C(u)$ and $C(v)$. For any $x \in C(u)$, the triangle inequality gives

$$\|x - p\|_2 \leq \|x - p_u\|_2 + \|p_u - p\|_2 \leq r_u + \|p_u - p\|_2. \quad (12)$$

The analogous inequality holds for $x \in C(v)$. Taking the maximum over the disjoint union $C(u) \cup C(v)$ yields exactly the recurrence above. Induction completes the proof.

Corollary 1 (directed vertex-set certificate). If the live clusters partition the original vertex set and every live radius is at most τ , then

$$\delta(V, V') \leq \tau. \quad (13)$$

Proof sketch. Each original vertex belongs to one live cluster and is within its representative radius. Its nearest output vertex is no farther than that representative. Taking the maximum over original vertices proves the bound. The reverse direction $\delta(V', V)$ is not implied and is therefore checked independently.

Rejecting a contraction when its propagated radius exceeds a safe threshold thus certifies the original-to-simplified direction for assigned representatives. The exported mesh is independently validated using a spatial nearest-neighbor structure. A safety margin below the official tolerance absorbs floating-point evaluation, rebasing, and output quantization.

This certificate is intentionally conservative. It can reject feasible contractions because it bounds every represented original vertex by a common ball. That conservatism is valuable during bulk simplification: the exact distance checker is reserved for checkpoints, while no accepted online contraction can silently accumulate unbounded displacement.

3.4 Perception-aware collapse objective

Pure QEM preserves supporting planes but does not directly model the flat-normal images. We use the composite proposal cost

$$E = E_Q + \lambda_N E_N + \lambda_C E_C + \lambda_R E_R. \quad (14)$$

The optional regularizer E_R represents case-specific curvature or projected-importance allocation. Heap insertion deliberately uses a cheaper approximation: it selects the radius-feasible position with minimum QEM and evaluates the composite objective only there. When the edge reaches the heap top, the current one-ring is rebuilt, every candidate position is scored once by the full composite, and valid positions are ordered by (composite cost, candidate type, coordinates). The readable core stores heap costs in double precision and uses a lexicographic edge tie-break. The byte-exact Accepted source used a more compressed float-key implementation; our reproducibility claim for that production artifact is its fetched-back hash and observed Kattis result, not universal cross-platform bit identity.

3.4.1 Projected-area normal distortion

For every surviving affected face f , let n_f and $n'_f(p)$ be its current and candidate normals.

Object-space area alone is a poor estimate of raster impact, because a large edge-on face may cover few pixels. We approximate total six-view coverage by a projected area A_f^{proj} and define

$$E_N(u, v, p) = \sum_{f \in A(u, v)} A_f^{proj} \left[1 - \text{clamp}(n_f \cdot n'_f(p), -1, 1) \right]^y. \quad (15)$$

The six axial cameras make the approximation inexpensive: projected components on the three coordinate planes, with orientation-aware signs and perspective depth, estimate raster influence. Alternative modes use object-space area or absolute projected components. This is not

a visibility term: it ignores occlusion, clipping, ownership, and foreground boundaries. Projected mode was most reliable on the Bunny-like case, though the specification-matching renderer still decided final operations.

Ignoring occlusion, clipping, and perspective, the unsigned axial projection sum is $A(|n_x| + |n_y| + |n_z|)$, which lies between A and $3^{1/2}A$. This elementary bound, derived in Appendix D, motivates a cheap importance surrogate but not an exact pixel count.

3.4.2 Cluster-normal memory

Repeatedly comparing a candidate with current faces creates reference drift. Every individual step can look acceptable while the aggregate surface slowly rotates away from the original. The implementation therefore defines the statistic from the original vertex-face incidence relation, not from an informal set of nearby faces. Let F_0 be the original faces and let

$$a_f = (p_{f,2} - p_{f,1}) \times (p_{f,3} - p_{f,1}) = 2 A_f n_f. \quad (16)$$

For every original vertex x , initialization adds a_f once for each incidence $x \in f$. If a live vertex v represents the multiset $C(v)$ of original vertices absorbed into it, its stored vector is

$$S_v = \sum_{x \in C(v)} \sum_{f \in F_0} 1[x \in f] a_f. \quad (17)$$

This incidence-weighted definition intentionally counts an original face once for each of its incident original vertices in the cluster. At a contraction, the stored values merge by $S_{u \cup v} = S_u + S_v$. Define $\hat{S}_v = S_v / \|S_v\|_2$ only when $\|S_v\|_2 > \epsilon_C$. For a surviving affected current face $g = (v_1, v_2, v_3)$, the old and proposed targets used by the code are

$$T_g = \sum_{j=1}^3 \hat{S}_{v_j}, T'_g = \sum_{j=1}^3 \hat{S}'_{v_j}, \quad (18)$$

where either contracted endpoint is replaced by the normalized merged vector in the proposed state. With b_g and b'_g denoting the current and proposed area vectors, and

$\phi(b, T) = [1 - \text{clamp}(b \cdot T / (\|b\|_2 \|T\|_2), -1, 1)]^\beta$, mode 2 implements

$$E_C(u, v, p) = \sum_{g \in A_{\text{surv}}(u, v)} \|b_g\|_2 [\phi(b'_g, T'_g) - \phi(b_g, T_g)]. \quad (19)$$

The subtraction is important: an operation that repairs accumulated drift can receive negative credit. If a stored vector or a three-vertex target cancels to norm at most ϵ_C , its normalized direction is undefined; the readable implementation rejects that candidate's cluster term rather than manufacturing a direction. On the 35,292-vertex case, the decisive early setting used $\lambda_N = 0.003$, $\gamma = 0.75$, projected area, $\lambda_C = 0.0001$, and $\beta = 0.5$.

Proposition 3 (merge-tree invariance for a fixed cluster partition). Fix the original incidences and a live cluster multiset $\mathcal{C}$. Any binary merge tree whose leaves are exactly the same elements of $\mathcal{C}$ produces the same stored vector

$$S_C = \sum_{x \in C} \sum_{f \in F_0} 1[x \in f] a_f. \quad (20)$$

Proof sketch. Every original vertex-incidence contribution is fixed at initialization. Internal nodes add the two disjoint child multisets, so associativity and commutativity of vector addition yield the same finite incidence sum. Normalization is performed only after the sum is requested and is not stored recursively.

This is deliberately a narrow order-invariance claim. Two collapse sequences that finish with different cluster memberships, surviving vertex identities, topology, or geometry need not have equal targets or penalties. Even for the same partition, floating-point addition can differ by roundoff unless the summation order is canonical. The property being protected is the mathematical original-support statistic, not the entire simplification trajectory.

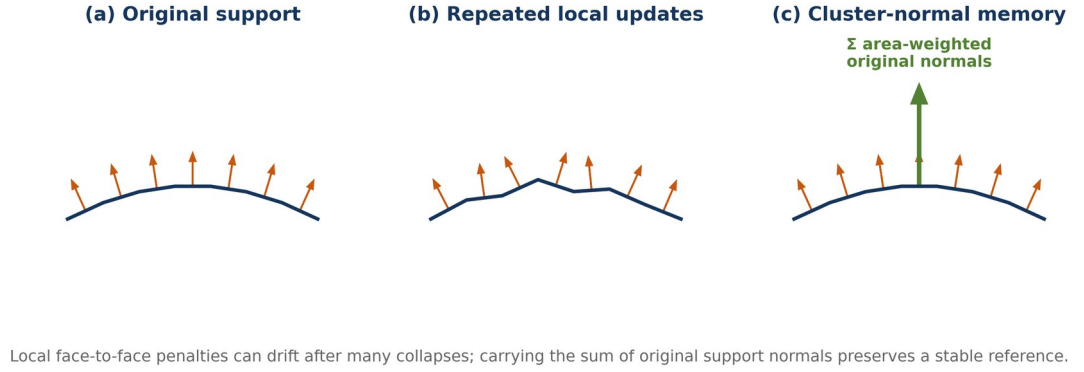


Figure 2. Cluster-normal memory retains original orientation evidence throughout a collapse tree.

3.5 Specification-matching local evaluator and component

diagnostics

The local CPU evaluator implements the public specification [1]: six positive and negative Cartesian cameras, camera distance 2.5, resolution 1024 by 1024, focal length 800, center at pixel coordinate 512, pixel-center sampling, nearest-triangle z-buffering, flat per-face normal encoding, perspective-correct reciprocal depth, 11-by-11 SSIM windows, foreground masking, and equal normal/depth weight. All release scripts pass resolution 1024 explicitly. Lower resolutions emit a screening-only warning because 512-square evaluations reversed several near-threshold rankings. All evaluator entry points parse meshes fail-closed and reject invalid indices, repeated face indices, zero-area faces, duplicate unoriented faces, non-finite coordinates, and trailing data instead of silently skipping them.

Three evaluators support diagnosis. The fast evaluator reports combined VPS. The component evaluator reports normal and depth SSIM for each view, identifying which signal is active. The oracle-normal variant replaces or isolates normal information to distinguish a bad topology from a bad normal surrogate. Their identity and invalid-input behavior are covered by synthetic tests. Agreement with the public statement is strong local evidence, but no local result is relabelled as an official hidden-test measurement.

3.6 Renderer-aware structural operators

3.6.1 Edge flips

Two adjacent triangles (a, b, c) and (b, a, d) can exchange diagonal ab for cd . The vertex count and vertex-set Hausdorff distance remain unchanged, but the piecewise-planar normal field can change substantially. We require positive orientation, no duplicate faces, preserved edge incidence, and unchanged closure before measuring the specification-matching render delta. Flips are especially valuable before deletion because they can redistribute normal error and make a one-ring easier to triangulate.

An edge flip leaves the output vertex set identical, so it leaves vertex count and both directed vertex-set distances unchanged. It can nevertheless change flat normals and z-buffer ownership; topology and raster effects therefore still require independent checks.

3.6.2 One-ring fan deletion and retriangulation

Deleting a vertex removes its incident fan and leaves a polygonal ring. For low valence, all combinatorially valid triangulations can be enumerated through Catalan recursion. For larger rings, restricted fans and dynamic-programming families reduce the search space. Each proposal is screened for orientation, duplicates, manifoldness, and distance before rendering. A deletion is accepted only when the locally measured worst relevant view remains above the chosen margin.

3.6.3 Coordinate and tangent fitting

After topology stabilizes, selected vertices are perturbed in coordinate, tangent, or normal directions. The objective is the local combined SSIM subject to triangle-area, orientation, and distance guards. Finite-difference or coordinate-search steps operate only inside bounded trust regions. Foreground cropping reduces cost on the million-vertex branch. Coordinate fitting is a margin-restoration tool, not a substitute for topology search: large gains generally required a flip or deletion first.

3.7 Discovering hidden constraints without hidden inputs

Kattis exposed only an overall verdict and score, not per-case render components or hidden meshes. We treated this as a black-box experimental-design problem. No hidden file was downloaded, reconstructed, or placed in the public artifact. The submitted program could, however,

compute statistics of its own input at runtime, and the displayed score could reveal a carefully controlled output count.

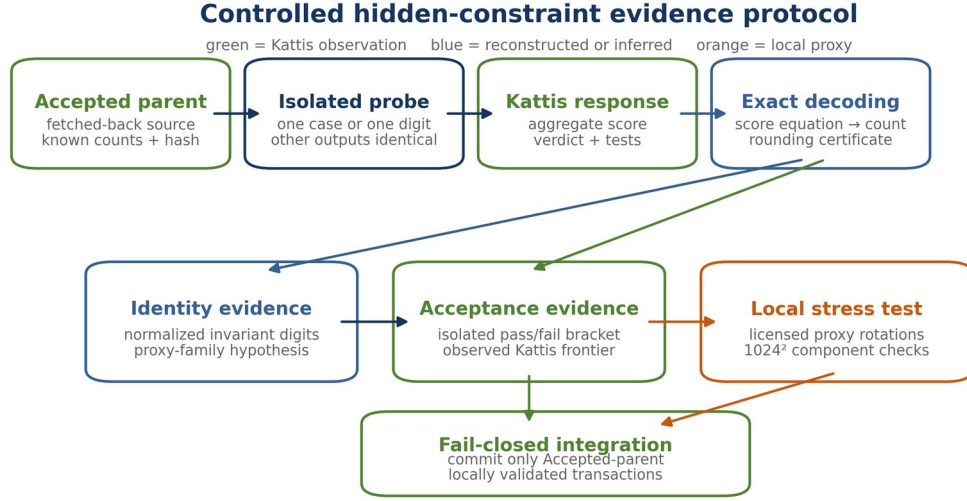


Figure 3. Controlled evidence protocol separating official observations, reconstructed or inferred quantities, and local proxy measurements.

3.7.1 Isolation and count decoding

A useful probe changes exactly one case k , while the other five outputs are byte-identical to a known Accepted parent. If the displayed score is S and all other output counts are known, the changed count is recovered as

$$|V'_k| = \text{round} \left(|V_k| \left[6 \left(1 - \frac{S}{100} \right) - \sum_{i \neq k} \frac{|V'_i|}{|V_i|} \right] \right). \quad (21)$$

For the 23,201-vertex case, one output vertex changes the score by approximately 0.00071836, so six displayed decimals uniquely determine the integer. This converts an aggregate score into a case-specific measurement while preserving scientific control: only one unknown changes.

If the displayed score error is at most ε , nearest-integer count decoding is unique whenever $2\varepsilon < 100 / (6|V_k|)$. With six displayed decimals, $\varepsilon \leq 0.5 \times 10^{-6}$; even the million-vertex branch has adjacent-count spacing about 1.65×10^{-5} , so the condition holds for all six cases. Appendix D gives the interval argument.

3.7.2 Exact-coordinate diagnostic carriers and scope rules

Let B_k be a conservative Accepted base count for case k , and let h be a small nonnegative diagnostic integer. A diagnostic branch outputs its normal manifold mesh plus unused vertices whose coordinates exactly duplicate coordinates already present in the face-referenced base mesh, giving

$$|V'_k| = B_k + h. \quad (22)$$

Faces continue to reference only the Accepted base geometry. The added count reduces the diagnostic score but does not endanger Kattis's stored best. After Kattis reports the score, the equation above recovers h . Multi-part fingerprints are emitted as base- B digits,

$$H = d_0 + B d_1 + B^2 d_2, \quad (23)$$

using separate isolated probes when the safe count slack cannot carry the entire integer.

Lemma 5 (invariance of an unreferenced exact-coordinate duplicate carrier). Let the base output have declared vertex list P , face list F , and face-referenced coordinate set X . Form P^+ by appending finite vertices, each with coordinates exactly equal to some element of X , while leaving F byte-identical. Under face-indexed surface semantics, the triangular surface, topology, normal/depth rasterization, and both directed vertex-set Hausdorff distances to the reference are unchanged. Only the declared count, vertex multiplicities, and predicates that inspect unused indices can change.

Proof sketch. Because F is unchanged, every indexed triangle and all surface-derived quantities are identical. As coordinate sets, $\{p : p \in P^+\} = \{p : p \in P\}$; adding a duplicate point

changes multiplicity but not the set. Nearest-neighbor minima and maxima in either Hausdorff direction are therefore unchanged. Exact duplication is essential: an arbitrary unused point leaves rasterization unchanged but can increase the output-to-reference directed Hausdorff distance. The repository's synthetic tests exercise both cases.

We used the carrier only under five fail-closed rules: coordinates had to be finite and exactly duplicated from a used base vertex; all faces had to remain byte-identical; the payload had to fit a precomputed nonnegative count slack; one isolated probe first established that the official parser permitted unused duplicates; and the diagnostic was never described as a quality improvement. The standalone validator reports unused and duplicate-coordinate vertices separately and offers a stricter `--require-all-used` mode so that the semantic assumption is visible rather than hidden.

The evidence boundary was equally strict. Statistics were computed solely by the submitted program from the mesh it was legitimately given; probes changed no external file, used no network or side channel, and attempted to recover only small decision-relevant invariants and counts. We neither reconstructed nor redistributed a hidden mesh. The method is an experimental-design technique for aggregate feedback, not a claim of access to organizer data.

The standard radix restriction $0 \leq d_j < B$ makes $H = \sum_j d_j B^j$ unique; Appendix D records the modular argument. Separate probes were still necessary to keep every carrier within independently validated count slack.

3.7.3 Rotation-invariant fingerprints

Proxy identity cannot be inferred reliably from vertex count alone. We used invariants computed from normalized geometry: quantized bounding-box ratios, genus and component count, sorted normalized edge-length multisets, edge-graph labels, orientation signatures, and order-sensitive hashes when raw ordering mattered. For the rotation-invariant length channel, define

$$L_E = \left(\frac{1}{|E|} \sum_{(a,b) \in E} \|p_a - p_b\|_2^2 \right)^{1/2}, \quad (24)$$

and quantize

$$q_e = \text{round} \left(B_q \frac{\|p_u - p_v\|_2}{L_E} \right), \quad (25)$$

then hashes the sorted multiset of q_e . Normalization removes translation and uniform scale; lengths remove rotation; sorting removes edge enumeration order. For the 23,201-vertex branch, three recovered base-15,776 digits were 2,382, 13,367, and 11,736, matching one exact-coordinate public proxy and differing from aggregated proxy variants. Independent pose, edge-graph, orientation, scale, and ordering codes agreed. This supplied evidence for selecting the relevant public proxy without exposing the hidden coordinates.

Under a similarity transform $p \mapsto s R p + t$, translation cancels in edge differences, orthogonal R preserves length, and division by L_E cancels s . Sorting removes enumeration order. Quantization adds only deterministic bin-boundary sensitivity; Appendix D gives the full algebra. Axis-aligned bounding-box ratios remain a separate pose-sensitive channel.

For the 35,292-vertex case, a conservative diagnostic output encoded normalized bounding-box statistics. The recovered values matched the public Bunny proxy, while acceptance behavior still showed a proxy-to-hidden renderer gap. We therefore separated **identity evidence** from **acceptance evidence**: a matching invariant justifies which geometry family to study, but only isolated Kattis probes establish an observed hidden-test boundary.

3.7.4 Frontier search and uncertainty control

Once a branch was identified, target counts were reduced by binary or bracketed search. Each probe changed one target or one operator family. A pass moved the observed Accepted frontier downward; a failure tightened the lower bound. Near runtime limits, repeated identical submissions distinguished stochastic load from systematic timeout, but repeated failures triggered engineering rather than indefinite resubmission.

Public proxies were rotated through sixteen orientations when practical. A candidate had to pass structural, distance, and specification-matching render checks across selected rotations, with an explicit margin. Rotation 04 was especially predictive for Lucy and Nefertiti during early

tuning, but no single rotation was treated as a decisive proxy. Hidden acceptance boundaries were recorded separately from local proxy thresholds to avoid retroactively labeling proxy scores as Kattis measurements.

3.8 Case-specialized solving

The score formula and different perceptual bottlenecks make one universal retained ratio suboptimal. Exact input size and conservative signatures choose a common core with specialized schedules.

Stage	Generic mechanism	Case-specific element	Runtime or offline
Parse and structural state	Buffered modified-OBJ parser; flat incidence arrays	Normalization only where required by a known branch	Runtime
Input recognition	Size plus fail-closed invariant checks	Expected signatures and branch dispatch	Runtime
Bulk simplification	Guarded QEM, candidate positions, radius and topology checks	Targets, weights, rebase schedule, heap caps	Runtime
Perceptual diagnostics	Shared rasterizer, component SSIM, topology and Hausdorff tools	Proxy rotations and safety margins	Offline
Structural search	Shared flip, fan-deletion, and coordinate-fit operators	Neighborhoods, accepted edit sequence, quantization scale	Offline
Deployment	Shared decoder, transaction checks, compact output	Bit-packed replay streams and reference caches	Runtime

Case	Input vertices	Final vertices	Retained	Dominant bottleneck and final strategy
Sphere-like	4,098	25	0.6101 %	Analytic smooth structure; guarded tiny branch
Armadillo	23,201	4,340	18.706 1%	Flat-normal detail; canonical replay, flips, fitted coordinates
Bunny-like	35,292	2,839	8.0443 %	Normal drift and silhouette; cluster memory, fan deletion, fitting

Case	Input vertices	Final vertices	Retain ed	Dominant bottleneck and final strategy
Lucy	49,987	3,030	6.0616 %	Tight normal/SSIM frontier; repeated flip- delete-fit cycles
Slender	377,084	7,400	1.9624 %	Curvature allocation and runtime; fitted replay, 40 flips, cache
Nefertiti	1,009,118	16,500	1.6351 %	Runtime, memory, facial detail; staged QEM and cropped fitting

The Sphere-like case admits a tiny guarded structural construction. Armadillo consumes the largest score loss and therefore received the most renderer-aware structural work. Bunny motivated cluster-normal memory. Lucy required careful operation ordering because local Hausdorff headroom did not imply normal-map headroom. Slender was geometrically compressible but exposed a runtime failure until reference caching. Nefertiti required staged passes, bounded heaps, and cropped fitting to stay within compute and memory limits.

3.9 Checkpoints, canonical replay, and transaction safety

Offline search can take hours; the submitted program cannot. We compile discoveries into deterministic operation streams. Vertex references are expressed in canonical current-mesh order or through stable local signatures rather than fragile temporary indices. Topology operators, fan choices, and quantized coordinate residuals are bit-packed. Repeated instruction fragments share dictionaries or base-94 payloads.

Replay is transactional. Before a specialized block, the current mesh is either checkpointed or reconstructible from a previously Accepted prefix. Every deletion, flip, and coordinate edit checks its expected local topology. A block commits only if its final count and structural invariants match; otherwise it rolls back or retains that checkpoint. This behavior makes the final source a compact guarded construction rather than an unchecked precomputed mesh dump.

Theorem 1 (fail-closed replay safety). Let $I(M)$ be the conjunction of all structural, count, checksum, and checkpoint invariants required of a replay state. Assume the initial checkpoint M_0 satisfies I , and every replay transaction either (i) commits a state M_{j+1} only after verifying $I(M_{j+1})$, or (ii) restores the preceding validated checkpoint. Then every externally visible checkpoint, including the released output, satisfies I .

Proof sketch. Induct on the transaction index. The base state satisfies I by assumption. For the inductive step, a successful transaction exposes a state only after its predicate is verified; a failed transaction exposes the previous validated state. Hence the next externally visible checkpoint satisfies I in either case. The conclusion follows for every finite prefix and the final state. This theorem covers only the predicates included in I ; perceptual acceptance and the complete Hausdorff check remain separate gates.

Corollary 3 (prefix-safe termination). If a timeout guard or unexpected local signature terminates specialization by returning the most recent validated checkpoint, the returned mesh still satisfies I . It may lose a prospective score improvement, but it cannot expose a partially applied operation block.

3.10 Runtime, memory, and source-budget engineering

Pointer-heavy half-edge structures were avoided on the largest input. Positions, faces, active flags, quadrics, cluster statistics, versions, and initial incidence live in flat arrays. New incidences are appended to compact linked storage. A binary heap stores candidate edges; endpoint versions invalidate stale entries lazily. Periodic pruning bounds heap growth. Input uses a buffered character parser and output uses a one-mebibyte writer.

Rebase checkpoints rebuild current adjacency and quadrics from the complete live arrays, controlling stale topology and allowing later stages to use smaller identifiers. Foreground crop boxes limit renderer work. Reference caches prevent repeated projection and normalization on Slender and Nefertiti. Static storage reduced allocator overhead and, on Kattis, improved runtime stability compared with otherwise equivalent dynamic variants.

Let V , F , and E denote the input vertex, face, and edge counts; H the total number of lazy-heap events; H_{max} the maximum resident heap size; K the number of exact rebases; R the number of replayed local operations; and d_{max} the maximum touched one-ring size. The implementation has the following parameterized bounds.

Stage	Time bound	Additional memory	Role
Parse, incidence, initial quadrics	$O(V + F + E)$	$O(V + F + E)$	Establish the first validated state
Lazy QEM search	$O(H \log H_{max})$	$O(H_{max})$	Rank and invalidate collapse proposals
K full rebases	$O(K(V + F + E))$	Reuses base arrays	Control drift and stale topology
Guarded local replay	$O(Rd_{max})$	$O(d_{max})$ scratch	Apply fitted flips, deletions, and moves

Summing the phase bounds gives online time $O((K+1)(V + F + E) + H \log H_{max} + Rd_{max})$ and memory $O(V + F + E + H_{max})$, excluding output buffering. The bound is deliberately parameterized because lazy invalidation can make H larger than E , while periodic pruning controls H_{max} .

The 131,072-byte source limit became a second optimization problem. Replay payloads were entropy-coded, recurring code fragments were factored, safe whitespace was removed, and unused diagnostic branches were stripped. Every minified candidate was compiled independently and run

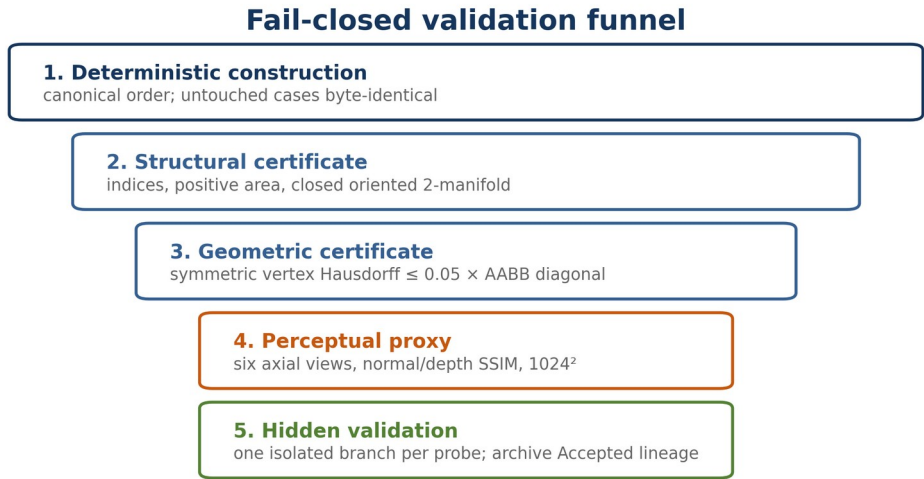
on the official sample. The final fetched-back source is 130,973 bytes, leaving 99 bytes of headroom.

Observable production quantity	Value	Evidence and interpretation
Fetched-back source	130,973 / 131,072 bytes	Official artifact size; 99-byte headroom
Physical source structure	453 lines; 450 macro definitions; 44 raw-string blocks	Reconstructed lexical counts; reflects minification, not algorithmic modularity
External gzip diagnostic	92,704 bytes	Reconstructed with gzip level 9; not used by Kattis and not a payload-size claim
Kattis tests	7 / 7	Official submission result
Kattis runtime field	N/A	The available submission endpoint returned an empty runtime field for 20082703
Kattis peak memory field	N/A	Not exposed by the archived manager output; no value is inferred

Acceptance establishes that the published time and memory limits were met on the organizer's environment, but it does not reveal the hidden per-case timing distribution. The 20082666-to-20082703 pair provides stronger architectural evidence than a fabricated runtime number: the uncached lineage completed six tests, whereas the cached release completed all seven.

3.11 Fail-closed validation funnel

Candidate validation proceeds from cheapest checks to the strongest external evidence. First, parse and index checks reject malformed meshes. Structural validation checks positive triangle area, duplicate faces, connectedness, oriented edge multiplicity, and the closed two-manifold condition. Next, an independent kd-tree search checks both directions of the stated vertex-Hausdorff distance. Only then does the specification-matching renderer compute normal, depth, per-view, and combined SSIM at an explicitly supplied resolution of 1024. Rotation suites test proxy robustness. Integration compares all untouched outputs byte-for-byte with the Accepted parent. Kattis is the last external gate.



A candidate is submitted only after every local gate passes; a runtime branch reverts if its transaction cannot be verified.

Figure 4. Fail-closed validation funnel used before an informative Kattis probe.

The funnel is conjunctive: a local release requires every independently implemented predicate to pass. This is a sound composition only for the explicit scopes in Table 2; it does not imply completeness, official evaluator parity, continuous-surface fidelity, or unseen-view quality.

The funnel is intentionally asymmetric: a candidate can be rejected for uncertainty even if it might pass. Near a hard threshold, false acceptance costs a submission and confounds diagnosis, whereas a conservative local rejection merely postpones a possible gain. This bias was appropriate because many branches had only a few thousandths of SSIM margin.

4. Results and Discussions

4.1 Submission result and exact score reconstruction

Submission 20082703 was reported by Kattis as Accepted, 7/7, score 93.830074. The fetched-back source is 130,973 bytes and has SHA-256 digest 9195d42a73a6b85c8ae30d731f532175bdcd7c2982d421143d631b4c64b1a92c. A time-stamped Kattis snapshot displayed rank 2 and, on the same page, the status “Unfinalized.” The artifact preserves those two fields verbatim and keeps them separate from the submission record [14,15].

Case	Origin al V	Final V'	Retained	Compression	Score loss
Sphere-like sample	4,098	25	0.610054%	99.389946%	0.101676
Armadillo	23,201	4,340	18.706090 %	81.293910%	3.117682
Bunny-like	35,292	2,839	8.044316%	91.955684%	1.340719

Case	Original V	Final V'	Retained	Compression	Score loss
Lucy	49,987	3,030	6.061576%	93.938424%	1.010263
Slender	377,084	7,400	1.962427%	98.037573%	0.327071
Nefertiti	1,009,118	16,500	1.635091%	98.364909%	0.272515
Global count aggregate	1,498,780	34,134	2.277452%	97.722548%	6.169926

Substitution into the official score definition gives

$$100 \left[1 - \frac{1}{6} \left(\frac{25}{4098} + \frac{4340}{23201} + \frac{2839}{35292} + \frac{3030}{49987} + \frac{7400}{377084} + \frac{16500}{1009118} \right) \right]. \quad (26)$$

The exact value is

$$93.83007422510956, \quad (27)$$

which rounds to the displayed score. The global compression ratio, $1 - 34134 / 1498780$, is 97.722548%; it is not the leaderboard formula, because the official mean gives every case equal weight [1].

The Accepted 7/7 verdict establishes that the submitted program passed the organizer's hidden suite, but Kattis did not expose per-case hidden normal, depth, or distance margins. Tables 7a and 7b therefore report only local public-proxy measurements retained with adequate provenance; unavailable component values are marked N/A rather than reconstructed from the aggregate.

Table 7a. Local proxy structural and finite vertex-set distance diagnostics.

Case / checkpoint	Topology	Reference to candidate	Candidate to reference	Hausdorff / tolerance
Armadillo 5,136	N/A	N/A	N/A	N/A
Armadillo 4,570	N/A	N/A	N/A	N/A
Lucy 3,030	Pass	N/A	N/A	0.935986
Slender 7,800	Pass	0.05286352	0.01112089	0.456384

Table 7b. Local proxy specification-matching perceptual diagnostics.

Case / checkpoint	Normal SSIM	Depth SSIM	Combined SSIM	Minimum-view combined
Armadillo 5,136	N/A	N/A	0.91701399	N/A

Case / checkpoint	Normal SSIM	Depth SSIM	Combined SSIM	Minimum-view combined
Armadillo 4,570	N/A	N/A	0.91722261	N/A
Lucy 3,030	N/A	N/A	0.90929170	N/A
Slender 7,800	0.87685919	0.94373269	0.91029594	0.90355001

The Slender row was recomputed with the released validator and evaluator at 1024 resolution; reference and candidate SHA-256 digests are recorded in `local_proxy_metrics.json`. The Armadillo and Lucy rows reproduce the scalar campaign records for checkpoints whose detailed component reports were not preserved. None of these rows is a hidden per-case metric, and the Slender 7,800 checkpoint is distinct from the released 7,400 output.

Submission 20082703 - reconstructed score 93.8300742251

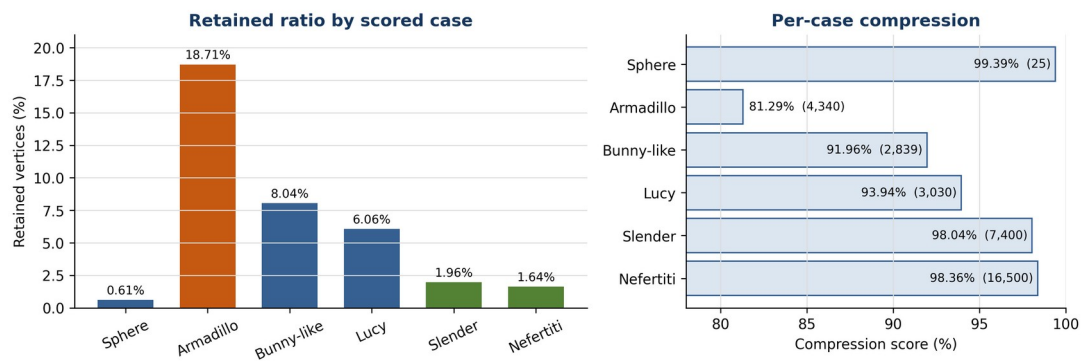


Figure 5. Retained ratios, compression, and per-case score contribution reconstructed for submission 20082703.

Figure 6 provides the missing qualitative counterpart to Table 7. It uses the evaluator's own PPM dumps for the lowest-scoring Slender proxy view: the reference and 7,800-vertex candidate are shown beside angular-normal and depth/silhouette error maps. All six panels share one crop and are generated without retouching. Green denotes low or zero error and magenta/red highlights larger disagreement. This diagnostic is a public-proxy measurement only; its purpose is to reveal where the scalar SSIM margin is spent, not to visualize a hidden test.

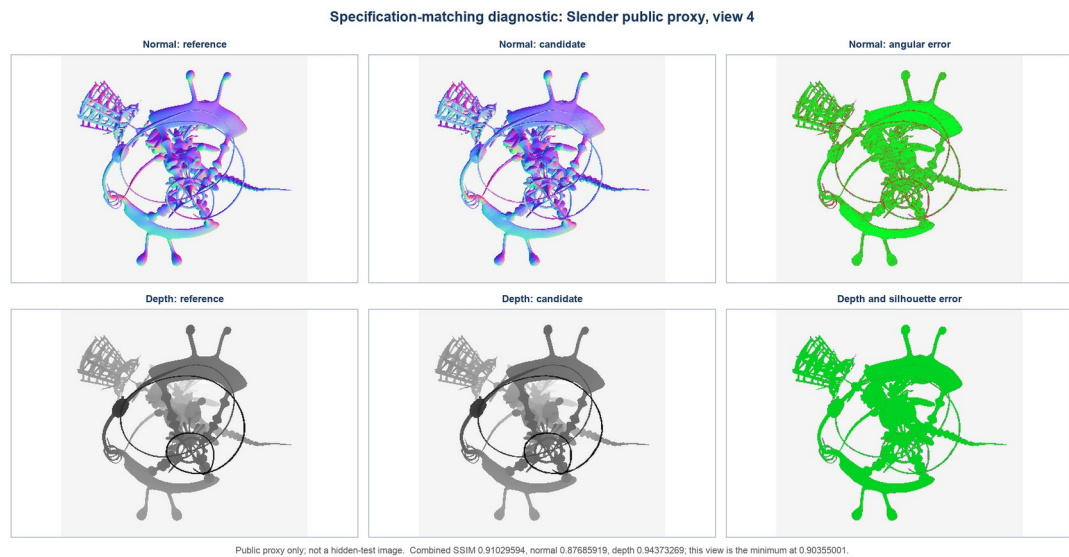


Figure 6. Reference, candidate, and error maps for Slender public proxy view 4 at 1024 square; this was the minimum view, combined SSIM 0.90355001.

4.2 Accepted progression

Development contained many small accepted steps and several structural jumps. The selected milestones below are official or archived high-waters; the public artifact contains the data used to generate the progression figure [14].

Submission or stage	Official score	Main change	Output-count evidence
Generic high-water	81.945906	Guarded generic simplification plateau	Archived lineage
Standalone QEM	86.998654	Case-routed robust QEM replaces legacy pipeline	Archived lineage
19932621	87.913148	Cluster-normal accumulation; Bunny-like 15.5%	Accepted 7/7
20020565 / 20021691 lineage	91.801173	Renderer-aware cycles and compressed replay	Accepted counts archived
20025898	92.932731	Armadillo, Bunny, Lucy, and Nefertiti checkpoints	[25, 5136, 3400, 3088, 7900, 17660]
20039214	93.600980	Renderer-optimized Armadillo at 4,570	Accepted 7/7
20051927	93.769981	Armadillo 4,340 and Bunny 2,848	Accepted 7/7

Submission or stage	Official score	Main change	Output-count evidence
20082128	93.812395	Bunny 2,839, Lucy 3,030, Nefertiti 16,500	Accepted 7/7
20082703	93.830074	Slender 7,400 with 40 flips and reference cache	[25, 4340, 2839, 3030, 7400, 16500]

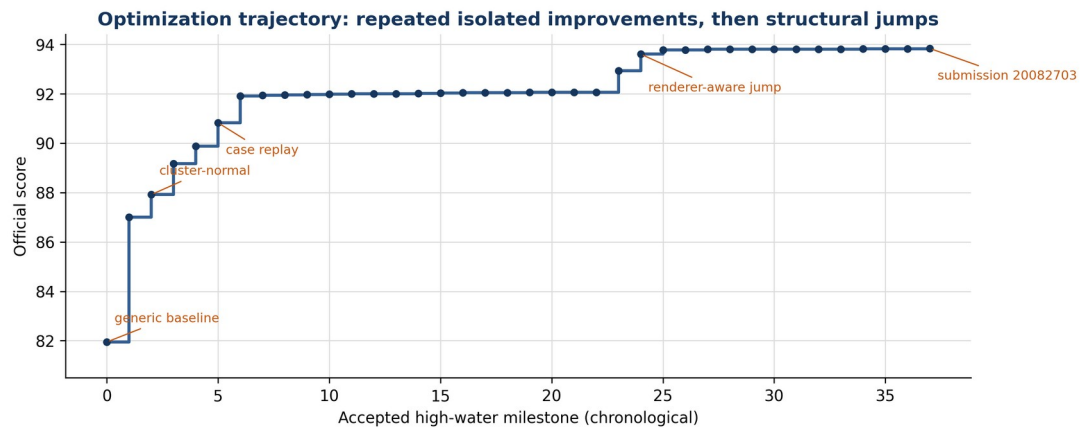


Figure 7. Accepted high-water trajectory from the generic plateau to submission 20082703.

The trajectory supports a central claim: small parameter tuning alone did not explain the release gain. The large jumps came from representation and workflow changes - cluster memory, proxy-family identification, structural replay, renderer-aware deletion, and replay compression. Later progress became incremental because each branch approached a different active constraint.

4.3 Hidden frontier reconstruction

The black-box protocol produced useful pass/fail boundaries before the later structural replays. These numbers are not direct hidden-mesh measurements; they are acceptance observations from isolated submissions.

Branch	Accepted frontier observation	Rejected neighboring observation	Interpretation
Armadillo, early QEM	34%, 33%, and 32% retained passed	31% retained failed	Pure QEM normal frontier near 31–32%
Bunny-like, early normal QEM	16% passed	Lower local-only settings failed	Current-face normal cost accumulated drift

Branch	Accepted frontier observation	Rejected neighboring observation	Interpretation
Bunny-like, cluster memory	15.5% passed	14.5% new-cost branch failed hidden	Cluster memory improved hidden frontier but proxy remained optimistic
Lucy, early QEM	9.5% passed	9.0% local and clustered variants failed	Hidden normal margin tighter than Hausdorff margin
Nefertiti, early QEM	4% passed	3% combined probe failed	Runtime and perceptual constraints both active

These boundaries guided allocation rather than defining the released outputs. Offline replay eventually moved far below the early QEM ratios: Armadillo to 18.706%, Bunny-like to 8.044%, Lucy to 6.062%, and Nefertiti to 1.635%. The comparison quantifies the value of structural and coordinate search beyond a better collapse weight.

The proxy audit also changed the research plan. Edge fingerprints showed that the exact 23,201-vertex source, not the agg3 or agg5 alternatives, matched the submitted input statistics. This eliminated a major source of proxy error and justified offline replay on the correct pose. The Bunny identity diagnostic matched the public family, yet its local rotations passed more aggressive counts than hidden Kattis. Identity and acceptance were therefore treated as separate uncertainties.

4.4 Case-by-case discussion

4.4.1 Sphere-like branch

The 4,098-vertex smooth input has a simple global shape and weak high-frequency detail. A guarded structural construction reaches 25 vertices while preserving closure and passing the release validation. Its small original size makes each retained vertex relatively valuable, but the branch was solved early and did not dominate later effort.

4.4.2 Armadillo

Armadillo contributes the largest release score loss because it retains 18.706% of its input. Early generic QEM passed around 32% and failed at 31%, demonstrating that flat-normal detail rather than the nominal Hausdorff bound was active. Input fingerprinting removed pose uncertainty. Canonical topology replay, low-valence deletion, edge flips, and tangent/coordinate fitting then reduced accepted counts through 5,136, 4,570, and 4,340. A 4,339 probe also passed, but the Slender integration was rebased on the byte-identical 4,340 lineage; the one-vertex difference is approximately 0.000718 points and was not worth destabilizing the release.

A notable local result was that the 4,570-vertex candidate slightly exceeded the combined VPS of a 5,136-vertex accepted reference despite using 566 fewer vertices. This is direct evidence that vertex count and perceptual quality are not monotonically coupled when topology and flat normals change together.

4.4.3 Bunny-like branch

Bunny motivated cluster-normal memory. Public rotations could pass lower ratios than hidden Kattis, but the local-current normal term became unreliable over long collapse chains. Carrying original support normals improved the Accepted high-water to 87.913148 at 15.5% retained. Later renderer-aware replay reduced accepted counts through 3,900, 3,850, 3,800, 3,775, 3,750, 3,400, 2,856, 2,848, and 2,839. The released topology is protected by checksum and fail-closed replay.

The decisive lesson is not that a larger normal weight is always better. Excessive weights froze locally visible faces and wasted vertices. The useful change was preserving the reference distribution across collapse history, allowing a modest normal term to remain meaningful late in simplification.

4.4.4 Lucy

Lucy exposed operation ordering. Direct 9% QEM and early clustered variants failed, whereas 9.5% passed. Subsequent renderer-ranked flips, coordinate fitting, and independent fan deletions advanced through 3,200, 3,150, 3,100, 3,097, 3,088, 3,040, and 3,030 vertices. The released 3,030 proxy measured combined VPS 0.90929170 and Hausdorff-to-tolerance ratio 0.935986. The remaining geometric headroom was larger than the remaining normal-map headroom, confirming that distance alone could not rank the last deletions.

4.4.5 Slender / Yeah Right

The 377,084-vertex model can be reduced aggressively, but generic QEM allocated triangles poorly around curvature. Plane quadrics were weighted by a mild curvature factor,

$$w_f = 1 + \lambda_\kappa \kappa_f^\eta, \lambda_\kappa = 2, \eta = 0.035, \quad (28)$$

followed by a fitted collapse replay and 40 topology flips. The 7,800 branch passed; submission 20082666 with the 7,400 geometry returned 6/7, and the campaign log attributes the missing test to runtime. A reference cache removed repeated work, and otherwise equivalent submission 20082703 passed 7/7. This is the cleanest systems ablation in the campaign: geometry was already locally valid, while cache architecture determined full-suite acceptance.

4.4.6 Nefertiti

The million-vertex input dominated compute and memory. The accepted schedule begins with a 2.2% QEM stage, rebuilds incidence and quadrics, applies several normal-sensitive pre-passes, and finishes with late passes plus bounded multi-view coordinate fitting. Foreground cropping and heap caps were essential. Accepted counts progressed through 18,669, 17,660, 17,100, 17,000,

16,900, 16,800, 16,700, 16,650, and 16,500. Lower geometrically valid candidates could fail SSIM or runtime. This branch shows why a feasible offline mesh and a deployable contest program are different objects.

4.5 Ablation evidence

Only comparisons with an otherwise stable lineage are treated as ablations. The matrix names what was held fixed and grades the causal interpretation; multi-variable milestone jumps are excluded.

Hypothesis / intervention	Held constant	Baseline	Intervention result	Inference, evidence, and strength
Original-support memory lowers long-horizon Bunny drift	Case target	Local-normal branch	Cluster-memory	Consistent with reduced drift; official frontier evidence, moderate , because code lineage was not a one-line toggle
	schedule and	passed at	branch	
	five other	16.0%;	passed at	
	branch outputs	14.5%	15.5%;	
	in the isolated lineage	revised-cost probe rejected	submission 19932621, 87.913148	
Renderer-aware structure can dominate count on Armadillo	Public proxy, evaluator, 1024 resolution	5,136 V; combined 0.91701399	4,570 V; combined 0.91722261	566 fewer vertices and +0.00020862 combined; local controlled pair, high for proxy , not hidden causality
First normal-fitting pass changes feasibility at fixed Nefertiti count	Output count 16,700 and integrated parent	16,700 V; rejected	16,700 V; Accepted, high-water 93.804841	Count cannot explain the verdict change; official pair, moderate , because hidden component margins are unavailable
Reference caching controls deployability of the 7,400 Slender geometry	Audited output geometry and replay payload	20082666; 6/7, campaign log reports runtime failure	20082703; Accepted 7/7	Runtime architecture determined full-suite acceptance; Kattis operational pair, high conditional on geometry-equivalence audit

Hypothesis / intervention	Held constant	Baseline	Intervention result	Inference, evidence, and strength
Slender count reduction	Other five output counts	7,800 V;	7,400 V;	Exactly +0.017679526754 from the formula;
explains the		93.81239469	93.83007422	reconstructed count
release score		8355	5110	causality, exact , while
delta				the cache is a separate runtime intervention

The Armadillo pair is a specification-matching local comparison rather than a Kattis quality measurement. The Nefertiti and cache rows are official isolated observations but do not expose the hidden normal/depth component. The final row is algebraically exact because the other five counts are identical. Larger milestone jumps, such as the transition to 92.932731, remain lineage evidence but are not labelled single-variable ablations because more than one specialized replay changed.

These experiments support three distinct causal claims. Additive cluster memory controls long-horizon reference drift; topology and fitting control perceptual feasibility at a fixed or lower count; and caching controls deployability without altering geometry. Keeping those claims separate prevents the score formula from being mistaken for a perceptual objective.

4.6 Negative results and what they taught us

Negative results shaped the final model and are as informative as accepted milestones.

- Fast generic libraries, MeshLab variants, meshoptimizer, and off-the-shelf simplifiers produced valid meshes but inferior flat-normal SSIM at aggressive ratios. Their objectives did not match the fixed renderer closely enough.
- Isotropic remeshing often preserved smooth depth and silhouettes but produced poor flat-shaded normal maps. Uniform triangle quality is not equivalent to renderer-optimal face orientation.
- Shared-vertex multilayer triangulations and alternative surface decompositions passed structural checks yet changed z-buffer ownership, causing SSIM loss.
- Low-resolution rendering was misleading near the threshold. Resolution 512 sometimes reversed candidate ordering relative to the official 1024, so it was demoted to screening only.
- Coordinate-only optimization produced small improvements but could not cross large plateaus. Topology had to change first; fitting restored margin afterward.
- Proxy success did not guarantee hidden acceptance. Sixteen public rotations reduced orientation overfitting, but isolated Kattis probes remained necessary.
- Heap growth, repeated incidence reconstruction, full-frame fitting, and redundant reference work caused timeouts on the largest cases. Heap pruning, crop bounds, static storage, and caching were necessary even when geometry was unchanged.

- Source minification introduced independent failures, including missing standard headers, altered storage duration, and accidental payload damage. Every release candidate was recompiled, sample-tested, byte-counted, and fetched back after acceptance.

4.7 Statistical and experimental interpretation

This was an optimization competition over six fixed instances, not a population study. We therefore do not report confidence intervals over independent meshes. The appropriate evidence is a traceable sequence of deterministic local measurements and official pass/fail observations. Multi-rotation proxy evaluation acts as a stress test, not as sampling from the hidden distribution. Kattis repetitions near runtime boundaries estimate operational stability but do not turn the benchmark into a stochastic trial.

The score is also an incomplete quality measure. It ranks vertex counts only after all hard constraints pass. Two accepted meshes with the same count receive the same score even if one has a larger SSIM margin. Consequently, specification-matching local VPS and distance ratios are used as safety diagnostics, while Kattis counts determine leaderboard value.

4.8 Threats to validity and limitations

Internal validity. Many late improvements combined code compression, runtime work, and geometry. We label only stable-parent comparisons as clean ablations. Hidden feedback was aggregate, so isolation and byte-identical unchanged outputs were required to identify the active case.

Construct validity. The official metric uses six views and flat normal/depth SSIM [1]. It does not measure arbitrary camera paths, materials, texture, or semantic plausibility. A mesh optimized for this evaluator may allocate triangles differently from a general interactive asset.

External validity. The released contest program specializes to fixed cases and embeds replay streams. It is not a universal drop-in simplifier. The reusable contributions are the guarded QEM core, cluster-normal memory, renderer calibration, structural search, fingerprint methodology, and fail-closed deployment workflow.

Geometric guarantee. The radius certificate is conservative and task-specific to vertex-set Hausdorff. Applications requiring continuous surface error should use sampling or envelope methods such as [7,8].

Black-box inference. Diagnostic fingerprints identify proxy families and parameters computed by the submitted program; they do not reveal the complete hidden input. A hash match can collide, so independent signatures and acceptance behavior are required. The public release excludes hidden inputs and third-party meshes.

4.9 Reproducibility and code availability

The complete curated artifact is available in the [public GitHub repository](#) [14]. It contains the byte-exact source fetched back from Kattis, machine-readable result metadata, the readable QEM and cluster-normal research core, evaluator sources, score reconstruction, this DOCX/PDF article, figures, build scripts, and a fail-closed release verifier. Official or hidden meshes are not redistributed.

The immutable release record is:

- Kattis submission: 20082703
- Judgement: Accepted, 7/7
- Displayed Kattis score: 93.830074
- Exact reconstructed score: 93.83007422510956
- Release output counts: [25, 4340, 2839, 3030, 7400, 16500]
- Source size: 130,973 bytes
- Source SHA-256:
9195d42a73a6b85c8ae30d731f532175bdcd7c2982d421143d631b4c64b1a92c

The release verifier reconstructs the score, checks source and article checksums, confirms the source limit, and compiles the fetched-back C++17 program. Researchers may build the readable simplifier and evaluator on any legally obtained triangular mesh; exact official integrity verification needs no contest mesh.

4.10 Model summary and operational workflow

The deployed system can be summarized as a compiler from a dense mesh to a validated perceptual replay:

- Parse and normalize the input; recognize the case with fail-closed signatures.
- Build quadrics, incidence, original normal summaries, cluster radii, and a versioned edge heap.
- Pop the cheapest candidate, test link, orientation, duplicate, normal, and radius conditions, then commit the contraction.
- Rebase at validated checkpoints to rebuild face-indexed adjacency and control numerical drift.
- Evaluate checkpoints with independent topology and distance tools plus 1024-square specification-matching component renderers.
- Search renderer-aware flips, fan deletions, and coordinate residuals in small neighborhoods.
- Canonicalize and compress accepted operations; replay them transactionally from an Accepted prefix.
- Cache large-case references, compact live vertices and faces, and stream the final OBJ output.
- Fetch every Accepted high-water, record counts and hashes, and use it as the only integration parent.

This workflow is neither purely online nor a hard-coded output. The online component guarantees scalability and a general safe structure. The offline component exploits the fixed evaluator. The replay component makes expensive discovery feasible under contest limits. The evidence protocol makes hidden feedback informative without pretending that proxy measurements are official.

5. Conclusion

The challenge was solved not by a single metric but by aligning every layer with the evaluator and deployment constraints. QEM supplied speed and geometric coherence [2]. Link and radius guards protected topology and distance. Projected normal cost and cluster-normal memory addressed the dominant perceptual failure of long collapse sequences. Specification-matching local rendering found structural and coordinate edits invisible to local geometry. Controlled black-box probes reduced uncertainty about hidden identities and acceptance frontiers. Compression, caching, and transactional replay converted offline discoveries into a 130,973-byte program that passed within 21 seconds.

The Accepted 7/7 result at displayed score 93.830074 and the exact count reconstruction support the hybrid strategy. The broader lesson is methodological: maintain a conservative invariant during large-scale optimization, measure the specification-matching rendered objective at carefully chosen checkpoints, isolate every external experiment, and deploy only deterministic operations backed by traceable evidence. Under a hard perceptual threshold, the strongest solver is not the one with the most elaborate local cost; it is the one that couples a scalable proposal model, a high-fidelity measurement model, and a fail-closed evidence and deployment pipeline.

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Appendix A. Core simplification pseudocode

```

load mesh M

build quadrics Q, original cluster normals S, and radius certificates r
build flat incidence arrays and a versioned candidate heap

while activeVertexCount exceeds checkpointTarget:
    (u, v) <- cheapest current edge candidate
    generate endpoint, midpoint, weighted, segment, and analytic positions
    order positions by QEM, projected-normal, and cluster-normal cost
    for position p in that order:
        if linkCondition(u, v) and validAffectedFaces(p)
        and normalGuard(p) and radiusCertificate(p) <= safeTolerance:
            commit contraction of u into v at p
            merge quadrics, original normal summaries, and radii
            refresh affected heap entries
            break

for each rendererOptimizedTransaction:
    decode flips, fan deletions, and coordinate residuals
    apply only when canonical local preconditions match
    commit only if expected count and structural checks pass
    otherwise retain the Accepted checkpoint

compact live vertices and faces
stream the OBJ output

```

Appendix B. Principal parameters in the release lineage

Branch	Base stage	Normal term	Additional mechanism	Offline tail
Sphere-like	Guarded structural reduction	Not required	Full structural checks	Target 25
Armadillo	Staged QEM checkpoints	$\lambda_N = .01$, $\gamma = .75$	Tangent and coordinate fitting	Canonical replay to 4,340
Bunny-like	Checkpoint near 8.043%	$\lambda_N = .003$, $\gamma = .75$	$\lambda_C = 10^{-4}$, $\beta = .5$	Flips, deletions, fit to 2,839
Lucy	7.448% base then replay	$\lambda_N = .001$, $\gamma = 1$	Repeated fit and mesh refresh	Deletion and flip streams to 3,030

Branch	Base stage	Normal term	Additional mechanism	Offline tail
Slender	2.2% curvature-aware QEM	Curvature allocation	$\lambda_k = 2$, $\eta = .035$	Fitted replay and 40 flips to 7,400
Nefertiti	2.2% first QEM stage	$\lambda_N = .0003$, $\gamma = .75$	Five pre-passes and two late passes	Cropped fitting to 16,500

Parameters were calibrated using specification-matching local rendering and frozen once a branch became part of an Accepted lineage. They are task-specific settings, not universal constants.

Appendix C. Verification checklist

- Compile the fetched-back source as C++17 and confirm that its size does not exceed 131,072 bytes.
- Verify the SHA-256 digest before treating the source as the release artifact.
- Run the official sample and confirm that the first output line is 8 12.
- Validate indices, positive triangle areas, duplicate faces, oriented edge multiplicity, connectivity, and closed two-manifold structure.
- Evaluate both directions of the official vertex-Hausdorff distance against five percent of the original bounding-box diagonal.
- Render all six views at 1024 by 1024; do not substitute 512 by 512 for a release decision.
- Record normal, depth, per-view, and combined SSIM rather than relying on one aggregate number.
- Archive and checksum each Accepted source; byte-compare unchanged cases with the integration parent.

Appendix D. Supporting derivations and proof details

Marginal score. Replacing $|V'_i|$ by $|V'_i| - 1$ changes only ρ_i by $-1/|V_i|$. Substitution into the affine score formula gives $+100/(6|V_i|)$. No claim follows if feasibility changes.

Axial projection bound. Orthographic projection onto the plane normal to axis j multiplies triangle area by $|n_j|$. Summation gives $A\|n\|_1$. Since a unit normal has $\|n\|_2 = 1$, the norm inequalities $1 \leq \|n\|_1 \leq 3^{1/2}$ give the stated interval. Perspective, clipping, and visibility are intentionally outside this derivation.

Rounded-count uniqueness. Exact scores for adjacent counts form a one-dimensional lattice with spacing $\Delta_k = 100/(6|V_k|)$. A displayed value with error at most ε corresponds to an interval of width 2ε . If that width is smaller than Δ_k , at most one lattice point lies in the interval.

Radix uniqueness. Assume two base- B digit sequences with digits in $[0, B-1]$ encode the same integer, and choose their lowest differing position j . After subtracting and dividing by B^j , the nonzero leading difference has magnitude below B , while all remaining terms are divisible by B , a contradiction modulo B .

Similarity-invariant edge channel. For $p' = sRp + t$, an edge difference becomes $sR(p_u - p_v)$, whose length is $s\|p_u - p_v\|_2$. The root-mean-square scale L_E is multiplied by the same s , so every normalized length is unchanged. Sorting removes enumeration order; deterministic quantization defines the finite stored channel.

Edge-flip vertex metric. A pure flip changes only two face triples. The output coordinate set is identical before and after the operation, so its cardinality and both nearest-neighbor vertex-set distances are identical. Face normals, visibility, and manifold orientation are not implied and remain separately checked.

Appendix E. Evidence ledger schema

Every selected experiment is represented as one JSON object in `paper/source/data/evidence_ledger.jsonl`. Required fields are `event_id`, `evidence_level`, `submission_id`, `parent_submission_id`, `changed_branch`, `intervention`, `verdict`, `official_score`, `output_counts`, `source_sha256`, `local_metrics`, and `interpretation`. Unknown values are JSON `null`, never prose such as “not exposed” inside a numeric column. Official, reconstructed, experimental, and inferred fields are kept distinct, and rejected probes are retained alongside Accepted milestones.